

Continuous Distribution.

Defn. A r.v. Y is said to be **continuous** if there exists a function

$$f_Y : \mathbb{R} \longrightarrow [0, \infty)$$

such that

$$\mathbb{P}[Y \in [a, b]] = \int_a^b f_Y(y) dy \quad \text{for all } a \leq b.$$

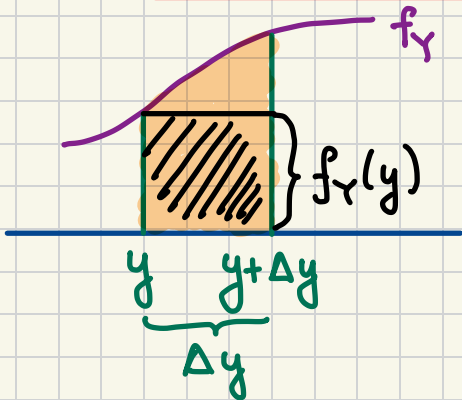
The f'ction f_Y is called the probability density function (pdf) of Y .

Properties:

- $f_Y(y) \geq 0$ for all y
- $\int_{-\infty}^{\infty} f_Y(y) dy = 1$

Note:

- For a pmf p_Y , we have that $p_Y(y) \leq 1$, for all $y \in S_Y$.
- For a pdf f_Y , it's possible that $f_Y(y) > 1$ for some y .



$$\mathbb{P}[Y \in [y, y + \Delta y]] = \text{shaded area} \approx f_Y(y) \cdot \Delta y \approx f_Y(y) dy, \text{ i.e.,}$$
$$\mathbb{P}[Y \in [a, b]] = \int_a^b f_Y(y) dy$$

Caveat: There are r.v.s that are neither discrete nor continuous!

Example. Y is uniformly distributed between l and r .

$$Y \sim U(l, r)$$

$$f_Y(y) = \begin{cases} \kappa & \text{for } y \in [l, r] \\ 0 & \text{for } y < l \text{ and } y > r \end{cases}$$

$$\int_l^r \kappa dy = 1 \Rightarrow \kappa = \frac{1}{r-l}$$

We introduce, for any subset $A \subseteq \mathbb{R}$,
 $\mathbb{1}_A : \mathbb{R} \longrightarrow \mathbb{R}$

$$\mathbb{1}_A(y) = \begin{cases} 1 & \text{if } y \in A \\ 0 & \text{if } y \notin A \end{cases}$$

This f'n is called the **indicator function**.

$$f_Y(y) = \frac{1}{r-l} \mathbb{1}_{[l, r]}(y)$$

M378K Introduction to Mathematical Statistics

Problem Set #5

Continuous distributions.

Problem 5.1. Source: Sample P exam, Problem #33.

The lifetime of a machine part has a continuous distribution on the interval $(0, 40)$ with probability density function f_X , where

$$f_X(x) \propto \frac{1}{(10+x)^2}$$

is proportional to

on the interval.

Calculate the probability that the lifetime of the machine part is less than 6.

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Density:

$$f_X(x) = \frac{25}{2} \cdot \frac{1}{(10+x)^2} \cdot \mathbb{1}_{(0,40)}(x)$$

$$\mathbb{P}[X < 6] = \int_0^6 f_X(x) dx$$

$$\frac{15}{32}$$

